

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Summer Examination – 2023****Course: B. Tech.      Branch : E&TC/ ECE / Electronics Engineering****Subject Code & Name: Signal & System (BTETC402/BTEXC402)      Semester : IV****Max Marks: 60****Date:15/07/2023****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                         |            | <b>12</b> |
| A) Sketch the following signal                                                                                                                                                                                                                                                                                                                                                                      | CO 01      | 6         |
| i) $x(t) = 3r(t) - 3u(t-3)$                                                                                                                                                                                                                                                                                                                                                                         |            |           |
| ii) $x[n] = \delta[n] + \frac{1}{2}\delta[n-1] + (\frac{1}{2})^2\delta[n-2] + (\frac{1}{2})^3\delta[n-3]$                                                                                                                                                                                                                                                                                           |            |           |
| B) Derive the expression for relationship between input and output of LTI system in time domain                                                                                                                                                                                                                                                                                                     | CO 02      | 6         |
| C) Determine if the following systems are time-invariant, linear, causal                                                                                                                                                                                                                                                                                                                            | CO 01      | 6         |
| i) $Y(t) = t*(X(t))$                                                                                                                                                                                                                                                                                                                                                                                |            |           |
| ii) $Y[n] = X[n] + n*X[n+1]$                                                                                                                                                                                                                                                                                                                                                                        |            |           |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                          |            | <b>12</b> |
| A) Explain the Even and odd signal with example. Also show that sum of even part and odd part of signal is equal to original signal.                                                                                                                                                                                                                                                                | CO 01      | 6         |
| B) Prove the following statements :<br>i) The convolution of an odd and even function is odd.<br>ii) The convolution of an even and even function is even.<br>iii) The convolution of an odd and odd function is odd.                                                                                                                                                                               | CO 02      | 6         |
| C) Find the Fourier series coefficients $C_k$ for the signal<br>$x(t) = (1/2) + (1/3)*\cos(t) + (1/2)*\cos(2t)$<br>Also find the FSC of following    i) $X(-t)$ ii) $X(t/2)$ iii) $x(t-2)$                                                                                                                                                                                                          | CO 03,04   | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                         |            | <b>12</b> |
| A) Let $x(n) = \delta(n+2) - \delta(n+1) + 2\delta(n) + \delta(n-1) - \delta(n-2)$ with discrete time Fourier transform $X(e^{j\omega})$ . Evaluate the following functions of $X(e^{j\omega})$ without computing the transform itself:<br>(i) $X(e^{j0})$ (ii) $X(e^{j\pi})$ (iii) $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$<br>(iv) $X(e^{j\omega})$ (v) Magnitude and phase of $X(e^{j\omega})$ | CO 03      | 6         |

|                                                                                                                                                                                                                                                                                                                                              |                 |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| <b>B)</b> If $X(e^{j\omega})$ is the Fourier Transform of a real sequence $x[n]$ then, show the following:<br>(i) $X(e^{j\omega})$ is conjugate symmetric<br>(ii) phase of $X(e^{j\omega})$ is anti-symmetric and Phase Spectrum is odd function<br>(iii) magnitude of $X(e^{j\omega})$ is symmetric and Magnitude Spectrum is even function | <b>CO 03</b>    | <b>6</b>  |
| <b>C)</b> Find Laplace transform of function: $f(t) = 4t^2 - 3\cos t + 5e^{-t}$ with $0 \leq t < \infty$ ;                                                                                                                                                                                                                                   | <b>CO 04</b>    | <b>6</b>  |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                   |                 | <b>12</b> |
| <b>A)</b> Perform the inverse Laplace transform of the following expression:<br>$F(s) = (3s + 7) / (s^2 - 2s - 3)$                                                                                                                                                                                                                           | <b>CO 04</b>    | <b>6</b>  |
| <b>B)</b> Determine the z transform and ROC for the signal<br>$x[n] = [3(2^n) - 4(3^n)] \cdot u(n)$                                                                                                                                                                                                                                          | <b>CO 04</b>    |           |
| <b>C)</b> Find Fourier transform of signal $x(t) = e^{at} \cdot u(-t)$ Also draw Amplitude and Phase Spectrum .                                                                                                                                                                                                                              | <b>CO 03</b>    | <b>6</b>  |
| <b>Q. 5 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                  |                 | <b>12</b> |
| <b>A)</b> Prove that energy in Continuous Domain and Energy in Fourier Domain remains Unchanged                                                                                                                                                                                                                                              | <b>CO 03</b>    | <b>6</b>  |
| <b>B)</b> Compute the inverse z-transform of signal $x(z) = (z+2) / (2z^2 - 7z + 3)$ if ROCs are<br>i) $ z  > 3$ ii) $ z  < 1/2$ iii) $1/2 <  z  < 3$                                                                                                                                                                                        | <b>CO 04</b>    | <b>6</b>  |
| <b>C)</b> Determine the DTFS of $x[n] = \{1, 2, 1, 0\}$ with period $N=4$ .                                                                                                                                                                                                                                                                  | <b>CO 03,04</b> | <b>6</b>  |

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